ESP32-Based Multi-Zone Automotive Petrol Fuel Vapor Monitoring and Leak Localization System

Embedded Systems Project

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Proposed Solution

To improve vehicle safety, we propose a **Smart Automotive Fuel Leak Localization and Alert System** that continuously monitors areas where fuel leaks are most likely to occur.

The system uses **three MQ-2 gas sensors** strategically placed near:

- **Engine Bay** – Detects vapors from leaking fuel injectors, fuel rail, or injector O-rings.
- **Fuel Line Area** – Detects leaks from cracked or damaged fuel lines.
- **Fuel Tank Area** – Detects vapors from the fuel tank, fuel pump, or fuel filter connections.

The **ESP32 microcontroller** continuously reads the gas concentration from all three sensors. It compares the readings against a predefined safety threshold.

Problem Statement

Problem

Fuel leaks in vehicles can occur due to:

- Damaged fuel tank
- Cracked fuel lines
- Worn fuel injector O-rings
- Faulty fuel pump or filter connections
- Loose or damaged fuel cap

These leaks release **flammable fuel vapors**, increasing the risk of fire, explosion, and vehicle damage.

Objective

Develop an embedded system that detects fuel vapors and has and adaptive threshold to manage the different levels of vaporization at different speeds

Proposed Solution

To improve vehicle safety, we propose a **Smart Automotive Fuel Leak Localization and Alert System** that continuously monitors the .

The system uses **MQ-2 gas sensor** strategically placed near:

- **Fuel Tank Area** – Detects vapors from the fuel tank, fuel pump, or fuel filter connections.

The **ESP32 microcontroller** continuously reads the gas concentration and identifies fuel leakage at different speeds of the vehicle

Working Principle

- If all readings are below the threshold:
  - OLED displays **"System Safe"**
  - **RGB LED glows Green**
  - Buzzer remains OFF
- If a moderate concentration of fuel vapor is detected:
  - OLED displays the **probable leak location**
  - **RGB LED changes to Yellow**
  - Buzzer beeps intermittently
- If a high concentration is detected:
  - OLED displays **"Danger! High Fuel Vapor Detected"**
  - Displays the **probable leak location**
  - **RGB LED changes to Red**
  - Buzzer sounds continuously

Concepts Applied

- Analog-to-Digital Conversion (ADC)
- Sensor interfacing
- GPIO control
- Threshold-based decision making
- Real-time monitoring
- I²C communication (OLED)
- Embedded programming using C/C++

Components

Component	Approx. Cost (₹)
ESP32	600
MQ-2 Sensors	600
OLED Display	250
Buzzer & LEDs	100
Breadboard & Wires	250
Miscellaneous	200
Total Estimated Cost	**≈ ₹2,000**

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Limitations

- MQ-2 detects combustible vapors — not gasoline-specific
- Probable zone estimate, not precise location
- Airflow & environment affect readings
- Prototype — not automotive-certified